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The story of Agni

India's intercontinental ballistic missile (ICBM) program has reached a new height. We look back on Agni V's journey and gaze into its future

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April 19th 2012: Wheeler Island, just off the coast of Orissa, was the testing site for the Defence Research and Development Organization or DRDO for the latest in its intercontinental ballistic missile technology, dubbed the Agni-V. It stands 17.5 meters high and 2 meters in diameters, with a weight of 50 tonnes and is capable of carrying warheads at a weight of 1,500 kilograms. The missile disappeared into the clouds at an altitude of 600 kilometres. The re-entry vehicle was then injected into the atmosphere at an altitude of 100 kilometres at a velocity of 6,000 meters per second. Armed with a dummy payload in its nose, it hit just a few meters away from its designated target. The approximate distance travelled by the Agni-V: 5,000 kilometres. The estimated distance it can travel is pegged to be around 5,800 kilometres (the distances of the Agni-III and IV were 3,000 kilometres and 3,800 kilometres, respectively), though in truth, it's still unknown. Even with two more validation tests to go, the missile was dubbed a success by authorities within the DRDO and the Indian media. The development of the Agni-V reportedly places India in a strong position amongst ICBM-capable nations.

Agni, India and the World

Agni is India's de facto missile system comprising of mid-range to intercontinental ballistic missiles or ICBMs. The latter is what fuelled concerns during the Cold War, as they are the primary means for delivering nuclear warheads across distances ranging from 5,000 to 10,000 kilometres. This is made possible by

launching them into low orbit, up to an altitude ranging from 150-400 kilometres, before crossing a sub-orbital flight path along the Earth's axis and re-entering the atmosphere at 100 kilometres to strike a target. They exist primarily as deterrents, though modern ICBMs have enough accuracy to hit even the smallest point in a city. The Agni I to IV are mid-range missiles, deployed by the Indian Army and Special Forces. With their success, the Agni family was eventually given special funding, which has led to the development of Agni V, India's first true ICBM.

Media coverage amongst different nations has been largely positive, with some criticism. "Objectively speaking, China does not spend much guarding against India, while India focuses a lot of attention on India," according to the Chinese state-run Global Times. The US



Launch of the Agni V missile from Wheeler Island

State Department complimented India's "solid non-proliferation record", with other authorities stating, "The lack of US condemnation of India's latest missile test demonstrates that the US is comfortable with Indian progress in the nuclear and missile fields and appreciates India's need to meet emerging strategic challenges."

Whatever the world thinks, it's obvious that the DRDO struck gold with its latest designs. Borne from a desire to guard against not one but two nuclear-equipped neighbours, the DRDO set to work to develop an appropriate follow-up to its Agni-III missile. "More than 80 per cent of the missile is indigenous, except for the electronic components which we import. Everything has been designed, developed and produced in our industry and our laboratories," said DRDO chief Dr. V.K. Saraswat.

ICBM Working and MIRV Advancements

Achieving the exact precision required for the Agni-V was one such task, which led to the development of a new propulsion system. Unlike its predecessor, the Agni-III, which follows a two-stage propulsion system, the Agni-V goes through three stages in its propulsion. The first stage is very similar to that of the Agni-III, which uses a solid-fuelled, advanced carbon composite as a booster. The second and third stage are modified, with the third stage being miniaturized and enlisting the help of composite motor casing to fly further ranges. The composites also help reduce the amount of weight on the missile.

One of the primary functions of the Agni-V, however, is mobility. It utilizes a canister, made of maraging steel, which preserves the missile as a tightly sealed atmosphere and ensuring its transportation for several years. A canister needs to be strong enough to withstand the external pressure and heat generated during a missile's launch and flight. Maraging steel, with a melting point of 1413 degrees centigrade and maximum tensile strength up to 2500 MPa, seems